If Statements & Comparisons

A **condition** is a True/False question; its indented block runs only when it is **True**.

1) What is a conditional statement?

- Write a colon : after the condition.
- Indent the statements inside the condition (normally four spaces).
- = assigns a value; == compares two values.

Demo 1 : condition is True

```
age = 20
if age >= 18:
    print("You are eligible to vote.")
print("Program finished.")
```

Output :

```
You are eligible to vote.
Program finished.
```

Demo 2 : condition is False

```
temperature = 25
if temperature > 35:
    print("It is very hot.")
print("Check complete.")
```

Output :

```
Check complete.
```

Demo 1: 20 >= 18 is True. The last print() is outside the block, so it always runs.

2) Comparison operators

Operator	Meaning	Example	Result
==	equal to	5 == 5	True
!=	not equal to	5 != 3	True
>	greater than	8 > 4	True
<	less than	2 < 6	True
>=	greater than or equal to	18 >= 18	True
<=	less than or equal to	10 <= 12	True

3) Compare two numbers

Example :

```
first_number = 15
second_number = 10
print(first_number == second_number)
print(first_number != second_number)
print(first_number > second_number)
print(first_number < second_number)
print(first_number >= 15)
print(second_number <= 10)
```

Output :

```
False
True
True
False
True
True
```

If-else & If-elif-else

Use **if-else** for two outcomes and **if-elif-else** for more. The first match runs.

1) The if-else statement

Demo 5 : even or odd

```
number = int(input("Enter an integer: "))
if number % 2 == 0:
    print("Even number")
else:
    print("Odd number")
```

Output :

```
Enter an integer: 7
Odd number
```

Demo 6 : pass or fail

```
marks = float(input("Marks out of 100: "))
if marks >= 35:
    print("Result: Pass")
else:
    print("Result: Fail")
```

Output :

```
Marks out of 100: 35
Result: Pass
```

% gives the remainder. Marks 34 gives Fail, 35 and 36 give Pass. >= includes 35.

2) The if-elif-else statement

Example :

```
marks = float(input("Enter marks out of 100: "))
if marks >= 90:
    print("Grade A")
elif marks >= 75:
    print("Grade B")
elif marks >= 60:
    print("Grade C")
elif marks >= 35:
    print("Grade D")
else:
    print("Fail")
```

Output :

```
Enter marks out of 100: 80
Grade B
```

Try these marks	95	80	65	40	20
Result	Grade A	Grade B	Grade C	Grade D	Fail

order matters: if marks >= 90 is True, Python skips the later conditions

Demo 8 : positive, negative or zero

```
number = int(input("Enter a number: "))
if number > 0:
    print("Positive")
elif number < 0:
    print("Negative")
else:
    print("Zero")
```

Output :

```
Enter a number: -4
Negative
```

Logical Operators & Nested If

Combine or reverse conditions with **and**, **or**, **not**, and put a check **inside** another check.

1) Logical operators

and: both True | or: at least one True | not: reverses True and False

A	B	A and B	A or B	not A
True	True	True	True	False
True	False	False	True	False
False	True	False	True	True
False	False	False	False	True

Demo 10 : and

```
age = int(input("Age: "))
has_id = input("ID card? (yes/no): ").strip().lower()
if age >= 18 and has_id == "yes":
    print("Entry permitted.")
else:
    print("Entry not permitted.")
```

Output :

```
Age: 20
ID card? (yes/no):   YES
Entry permitted.
```

Demo 11 : or

```
student = input("Student? (yes/no): ").strip().lower()
senior = input("Senior? (yes/no): ").strip().lower()
if student == "yes" or senior == "yes":
    print("You qualify for a discount.")
else:
    print("Standard price applies.")
```

Output :

```
Student? (yes/no): no
Senior? (yes/no): yes
You qualify for a discount.
```

.strip() removes outer spaces and .lower() makes lowercase, so YES, Yes and yes match

2) Nested if statements

A nested if is an `if` inside another `if`. The second check happens only after the first is satisfied. Each `else` belongs to its nearest `if`.

Example :

```
username = input("Enter username: ")
password = input("Enter password: ")
if username == "student":
    if password == "python123":
        print("Login successful.")
    else:
        print("Incorrect password.")
else:
    print("Username not found.")
```

Output :

```
Enter username: student
Enter password: python
Incorrect password.
```

classroom demo only: real applications never store passwords as plain text

Mistakes & Practice

Learn the seven classic mistakes, then try four tasks first. Compare with the model solutions.

1) Common beginner mistakes

Mistake	Wrong	Right
= instead of ==	`if age = 18:`	`if age == 18:`
forgetting the colon	`if age >= 18`	`if age >= 18:`
wrong indentation	body not indented	indent the body the same each time
separate ifs, not elif	all matching ifs run	if-elif-else runs only the first match
boundary conditions	> when 18 should pass	decide: > or >= ?
text vs numbers	`input() >= 18`	`int(input()) >= 18`
uppercase and spaces	`answer == "yes"` for " YES "	`answer.strip().lower() == "yes"`

2) Practice tasks and solutions

Task 1 Adult or minor.

Solution :

```
age = int(input("Age: "))
if age >= 18:
    print("Adult")
else:
    print("Minor")
```

```
Age: 17
Minor
```

Task 2 Pass or fail (40).

Solution :

```
marks = float(input("Marks: "))
if marks >= 40:
    print("Pass")
else:
    print("Fail")
```

```
Marks: 40
Pass
```

Task 3 Divisible by 3 and 5.

Solution :

```
n = int(input("Number: "))
if n % 3 == 0 and n % 5 == 0:
    print("Divisible by both")
else:
    print("Not both")
```

```
Number: 30
Divisible by both
```

Task 4 Free delivery (500).

Solution :

```
amount = float(input("Amount: "))
if amount >= 500:
    print("Free delivery")
else:
    print("Delivery charge applies")
```

```
Amount: 450
Delivery charge applies
```

Quick check

What does == do? **It compares two values** (= assigns). What does % return? **The remainder**.